Introduction to Machine Learning

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Contents

1 Introduction **2**
1.1 Classification 2
1.2 Training, validation, and testing 2
1.3 Overfitting and underfitting 3

Lecture 1: Introduction

Machine learning uses data to find patterns and make predictions. A model describes possible patterns; a learning algorithm uses data to select one. The amount and quality of data affect what can be learned.

1.1 Classification

Classification predicts a class label y from a feature vector x, using labeled examples. A decision boundary separates regions assigned different labels. In two dimensions, a linear classifier uses a line; in higher dimensions, a hyperplane.

Example: A lender can represent an applicant by income and credit card balance, then predict whether the applicant will default. Each training point has these two features and an observed label.

In k-nearest neighbors, find the k closest training points to a new input and predict their majority label. When $k = 1$, use the nearest point's label. The choice of model changes the boundary learned from the data.

An image can be represented as a vector of pixel intensities. For example, a 16-pixel image is a point in 16-dimensional space; a classifier can use labeled images to predict whether a new image is a seven.

1.2 Training, validation, and testing

The goal is to generalize: predict well on examples not used to fit the model. Partition labeled data into distinct sets:

1. Training set: fit each candidate model's parameters.
2. Validation set: compare trained models and select settings such as k; do not use it to fit their parameters.
3. Test set: evaluate the final selection on new examples. An evaluator may withhold the labels.

Classification error is the fraction of examples assigned the wrong label. Training error measures fit; validation error supports selection; test error assesses the final choice. The term "training data" can also mean all available labeled data, so check the context.

1.3 Overfitting and underfitting

A HYPERPARAMETER controls a learning procedure or model family. The number k is one example: small k can produce a detailed boundary, while large k averages more neighbors and tends to smooth it.

- OVERFITTING: test or validation error worsens because the model follows noise, outliers, or accidental patterns too closely. An OUTLIER is an atypical example, such as a wealthy borrower who defaults.
- UNDERFITTING: error worsens because the model is too inflexible to capture important patterns. A very large k can smooth away real local differences.

Example: With unique, consistently labeled training points, 1-nearest neighbors has zero training error: each point is its own nearest neighbor. This does not guarantee good predictions on unseen points. Validation helps choose k; repeatedly choosing models using test results can favor a lucky model, so retain a separate final test set.

Source. Jonathan Shewchuk, Introduction to Machine Learning, Spring 2025, Lecture 1.